

Case Study: Engineering High-Efficiency, High-Torque Solutions for Compact Electric Motors

Client: Manufacturers of e-Mobility, Precision Robotics, and Household Appliances

Prepared by: FEAAM GmbH

Subject: Maximizing Power Density and Efficiency in Compact Electric Motors

Executive Summary

In the market for small-scale electric motors, manufacturers face extreme pressure to deliver high performance within tight space and cost constraints. FEAAM GmbH has developed a suite of innovative technologies—including flux-barrier stators, double-rotor designs, and hybrid stators—that significantly enhance power density and efficiency. These solutions allow manufacturers to maximize motor performance without increasing the physical footprint or raw material costs. Toward the end of the design process, FEAAM can help by performing feasibility studies and 3D-numerical optimizations to identify the highest ROI for a specific product line.

The Challenge: The "Small Motor" Performance Bottleneck

Small motor applications in household appliances, power tools, and robotics often have rigid packaging requirements that limit motor size. Traditional motor designs struggle with several bottlenecks:

- **Wasted Space:** Standard designs have "dead zones" in the end-windings that consume volume but do not contribute to torque.
- **Magnetic Losses:** Concentrated windings create magnetic "noise" (harmonics) that generate heat and reduce efficiency.
- **Thermal Limits:** Small, enclosed motors can easily overheat, forcing a reduction in power to maintain safety.

The FEAAM Innovation: Multi-Faceted Design Optimization

FEAAM utilizes a proprietary toolset to optimize the internal architecture of small motors:

- **Flux-Barrier Stator:** Strategically placed slits in the stator teeth "filter" the magnetic field. This reduces harmful harmonics and boosts the useful magnetic wave, increasing torque density by approximately 20%.

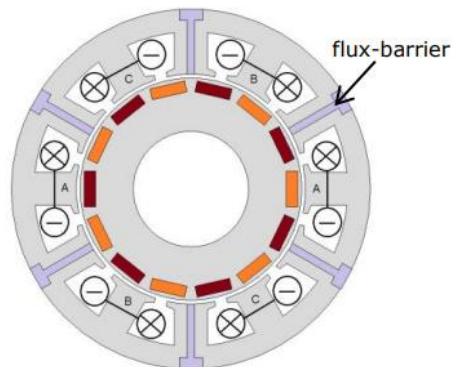


Fig. 1. Geometry of the 12-teeth/14-poles PM machine with the new stator structure.

- **Double-Rotor "Flat" Design:** For applications where length is strictly restricted (down to 10mm), a flat motor with rotors on both sides of the stator uses 100% of the available space for torque generation.

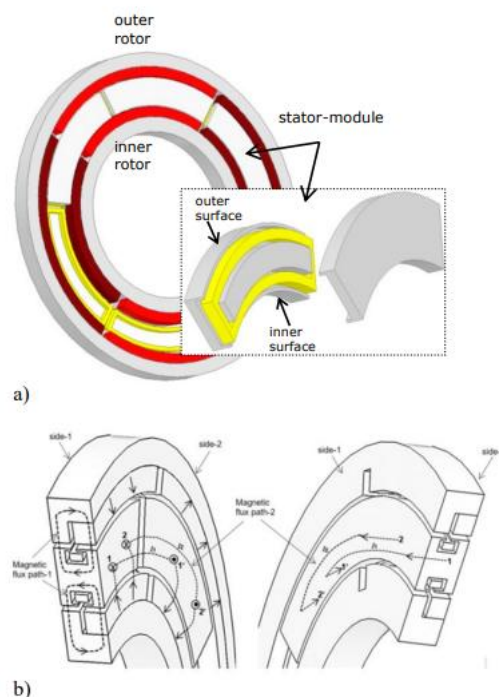


Fig. 2. The flat motor design with dual rotor. a) Motor configuration including stator module, coils, magnets and rotor core. b) Illustration of flux lines magnetic flux path.

- **Hybrid Stator with Extended Teeth:** This design uses extended tooth tips and Soft Magnetic Composite (SMC) materials to capture magnetic flux from the end-windings—areas usually wasted in standard motors—to further boost power.

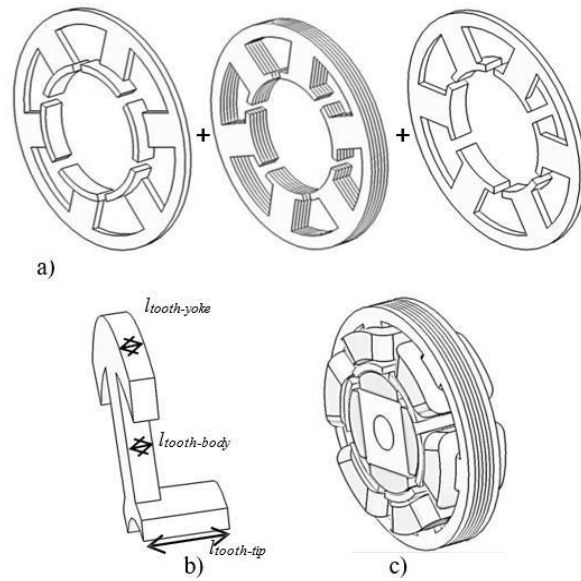


Fig. 3. a) Hybrid Stator components in exploded view. b) Tooth module of the stator end components. c) Four pole permanent magnet machine with hybrid stator and axially extended rotor.

Performance Validation:

Measured Results FEAAM has validated these technologies through rigorous numerical analysis and experimental testing across various high-growth sectors:

- **Ultra-Flat Robotics:** For precision robotic joints capped at 10mm axial space, FEAAM's double-rotor technology achieved a staggering 97% efficiency and a rated power of 274W.
- **High-Efficiency Fans:** By integrating the flux-barrier stator, FEAAM reduced motor size while achieving over 90% efficiency within a compact 45mm diameter.
- **Harmonic Filtering:** Empirical tests confirm a 73% reduction in the problematic first subharmonic and a 16% boost in the working harmonic, resulting in much cooler and quieter operation.
- **Traction Gains:** E-bike prototypes using this architecture showed a 15% torque increase while requiring lower excitation currents to reach peak power.

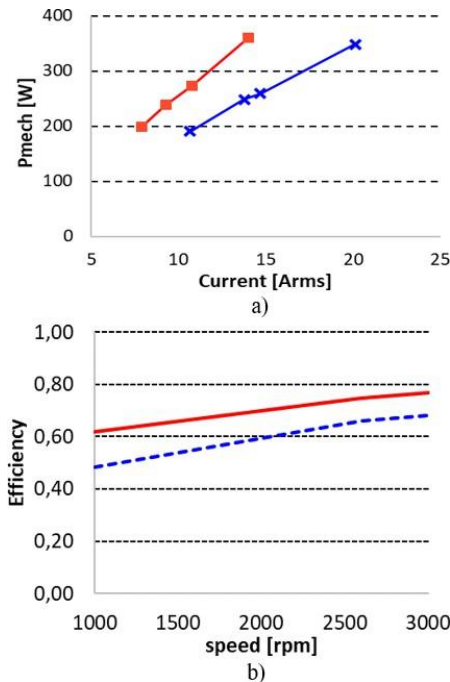


Fig. 4. Comparison of experimental results. In red for the flux-barrier motor, in blue for the reference motor. a) Mechanical power at 3000rpm. b) The e-drive efficiency (motor + gearbox) at maximal torque.

The Bottom Line for Business

FEAAM's specialized design techniques allow manufacturers to break through the physical limitations of small motor design:

- **Lower Total Costs:** Achieve higher performance using standard materials through superior electromagnetic architecture.
- **Sleeker Product Packaging:** Deliver more powerful tools and appliances without increasing their size or weight.
- **Market Differentiation:** Offer products that run cooler, quieter, and more efficiently (up to 97% efficiency) than competitors using "off-the-shelf" designs.

FEAAM can assist your team by performing feasibility studies and 3D-numerical optimizations for your specific product line, helping you break through physical packaging limitations to deliver superior performance without increasing motor size or raw material costs.